



RAINFALL — FULL MIND MAP

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1. SW Monsoon — India's Main Rainfall Source

- India ko **maximum rainfall South-West Monsoon** se milti hai — April-May mein Indian subcontinent ke north mein intense low pressure banta hai, ITCZ north ki taraf shift hoti hai, aur SW monsoon **do rain-bearing systems (Bay of Bengal branch + Arabian Sea branch)** mein India mein enter karta hai. [Q1]

2. NE Monsoon — Tamil Nadu's Main Rainfall

- Tamil Nadu apni annual rainfall ka **40-50% North-East Monsoon (October-December)** se paata hai — **26 December 2004 ko isi season mein Tsunami aayi thi.** SW monsoon mein **Chennai sabse kam rainfall receive karta hai** (Chennai ka main rainy season NE monsoon hai). [Q2, Q15, Q29]

3. Highest Average Annual Rainfall State (Sikkim/Meghalaya)

- Jab question pucha gaya tha tab **Sikkim (3300.7mm)** ko highest maana jaata tha — 2022 IMD data ke mutabik ab **Meghalaya (4216.1mm)** sabse zyada rainfall wala state hai. [Q3]

4. Same-Latitude Rainfall Contrast (Ajmer-Shillong)

- Same latitude (26°27' N) par hone ke bawajood, **Ajmer (623.5mm/year)** aur **Shillong (8084.7mm/year)** ki rainfall mein sabse zyada spashta antar hai — Bengaluru-Chennai, Mumbai-Vishakhapatnam, Nagpur-Kolkata jaisi comparisons se zyada. [Q4]

5. Meteorological Station Identification (Chennai)

- 26°C average annual temperature, 63cm rainfall, 9°C annual range — ye **Chennai** ki characteristics hain (winter 24°C, summer 30°C, annual range 8-9°C, average rainfall 60-70cm). [Q5]

6. City Rainfall Ranking (Kochi > Kolkata > Patna > Delhi)

- 2022 ke data ke mutabik descending rainfall order: **Kochi (3415.8mm)** > **Kolkata (1325.1mm)** > **Patna (736.2mm)** > **Delhi (675.9mm)** — India ki sabse zyada rainfall Mawsynram (Meghalaya) mein hoti hai. [Q6]

7. Mango Shower, Blossom Shower, Norwesters, Loo

- Local storm match: **Mango Shower** — Karnataka coastal areas (pre-monsoon, aam pakane mein madad karti hai), **Blossom Shower** — Kerala (coffee phoolon ko khilne mein madad), **Norwesters (Kal-Baisakhi)** — West Bengal/Assam (Baisakh mahine mein, tea-jute-rice ke liye faydemand), **Loo** — Punjab se Bihar (garam-shushk local winds). [Q7, Q8, Q9, Q10]

8. Lowest Rainfall Places (Leh)

- India mein **Leh** mein sabse kam rainfall (63.8mm/year) hoti hai — Jaisalmer Rajasthan ka lowest hai (372.9mm), Bikaner 474mm. [Q11, Q12]

9. Cherrapunji — Location & Rainfall Record

- Cherrapunji (ab Sohra) Meghalaya ke East Khasi Hills district (Khasi Hills) mein hai — Mawsynram ke baad dusra highest-rainfall record iske naam hai. [Q13, Q28]

10. Highest Rainfall Regions of India

- India mein sabse zyada rainfall **Western Ghats, Eastern Himalayan region** aur **Meghalaya** mein hoti hai — 250cm+ average annual rainfall. [Q14]

11. 'Thirsty Land' Paradox — Why India Lacks Water Despite Rain

- India ki 1257mm actual annual rainfall hone ke bawajood ise 'thirsty land' kaha jaata hai kyunki: (1) **paani ka rapid runoff**, (2) **rainwater ka quick evaporation**, (3) **sirf kuch mahinon (June-Sept) mein hi rainfall concentrate hoti hai** — teeno reasons combined. [Q15]

12. Rainwater Harvesting — Cost-Effective Strategy

- India ki **cost-effective water management strategy** — **Rainwater Harvesting** hai (embankments-tanks banake) — **Tamil Nadu** isme leading state hai. [Q16]

13. 200mm Average Rainfall Zone (Zanskar & W. Rajasthan)

- ~200mm average annual rainfall **Zaskar range (northern J&K/Ladakh) aur western Rajasthan** mein hoti hai. [Q17]

14. Jharkhand's Monsoon — South-West

- Jharkhand ki rainfall **South-West Monsoon (June-September)** se hoti hai. [Q18]

15. 100-150cm Rainfall Shaded Map Zone

- Diye gaye India map ke shaded region mein mean annual rainfall **100 se 150cm (broadly 100-200cm range)** ke beech hoti hai. [Q19]

16. Pushkar Hills Rainfall — Balotra Floods

- Pushkar Hills mein bhaari rainfall hone par **Balotra (Rajasthan) mein baadh** aati hai. [Q20]

17. 'Rainy Day' Definition — 2.5mm Threshold

- IMD ke mutabik 'rainy day' wo din hota hai jab **24 ghanton mein 2.5mm ya usse zyada rainfall** ho. [Q21]

18. Ganga Plain Rainfall Gradient (Assertion-Reason)

- Ganga plain mein **west/north-west ki taraf jaane par monsoonal rainfall kam hoti hai** (Punjab plains ~600mm, Gangetic plain ~1500mm) — kyunki **moisture-bearing monsoonal winds jaise-jaise upar (west) badhti hain, unki humidity ghat jaati hai** (Assertion sahi, Reason sahi explanation hai). [Q22]

19. All India Water Divide Map

- Diya gaya India ka map '**All India Water Divide**' dikhata hai — Isohyet, national highways, ya major highway routes NAHI. [Q23]

20. Climate Reason-Match Set (Latitude, Altitude, Depressions)

- Climate-reason match: **Chennai Kolkata se garam hai — latitude ka difference; Himalaya mein snowfall — altitude; West Bengal se Punjab tak rainfall ghatti hai — sea se distance; Sutlej-Ganga plain mein winter mein kuch rain — western depressions.** [Q24]

21. Watershed Development in Semi-Arid Regions

- India ke semi-arid regions mein watershed development ki pehchaan **seasonal rivers ka paani embank karke tanks ki system banana** hai — ~120,000 chhote tanks se ~4.12 million hectares zameen ki sinchayi hoti hai. [Q25]

22. Desert Development Programme — District Count

- Desert Development Programme (DDP, 1977-78 mein shuru) us waqt **5 states ke 21 districts (131 blocks) mein tha** — abhi **7 states ke 40 districts (235 blocks)** cover karta hai. [Q26]

23. Inland Waterways Underdevelopment (Assertion-Reason)

- India mein **inland waterways** acchi tarah se develop **NAHI hain** (Assertion sahi) kyunki **saal ke sirf 4 mahine hi rainfall** hoti hai, isliye rivers mein paani ki kami rehti hai (Reason sahi explanation hai). [Q27]

24. Cherrapunji — Khasi Hills Location

- Cherrapunji **Khasi Hills, Meghalaya** mein hai — Naga, Garo, Mikir Hills mein **NAHI**. [Q28]

25. Chennai's Low SW-Monsoon Rainfall

- SW monsoon (1 June-30 September active) se **Chennai sabse kam rainfall receive karta hai** (Kolkata, Mangalore, Delhi se kam) — Chennai ki main rainfall NE monsoon se aati hai. [Q29]